

SAT Hard Question 6

Jack Zeng

May 21, 2026

Novel Prep

Question 1

Given the expression

$$\sqrt[5]{70n \left(\sqrt[5]{70n} \right)^2}$$

for what value of x is the given expression equivalent to $(70n)^{30x}$, where $n > 1$?

Answer to Question 1

Correct Answer: $\frac{3}{350}$

Question 2

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as

$$(hx + k)(x + j)$$

where h , k , and j are integer constants.

Which of the following must be an integer?

- A. $\frac{b}{h}$
- B. $\frac{b}{k}$
- C. $\frac{45}{h}$
- D. $\frac{45}{k}$

Answer to Question 2

Correct Answer: D

Question 3

The expression

$$6\sqrt[5]{3x^{45}} \cdot \sqrt[8]{8x}$$

is equivalent to ax^b , where a and b are positive constants and $x > 1$.
What is the value of b ?

Answer to Question 3

Correct Answer: $\frac{73}{8}$

Question 4

In the given equation, c is a positive constant.
Solve for x :

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

Which of the following is one of the solutions to the given equation?

- A. $-c$
- B. $-c^2 - 39^2$
- C. $-\sqrt{39^2 - c^2}$
- D. $\sqrt{c^2 + 39^2}$

Answer to Question 4

Correct Answer: D

Question 5

The quadratic function is given as

$$f(x) = ax^2 + 4x + c$$

where a and c are constants.

The graph of $y = f(x)$ is a parabola that opens upward and has a vertex at (h, k) , where h and k are constants.

If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

- I. $c < 0$
 - II. $a \geq 1$
- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Answer to Question 5

Correct Answer: D

Question 6

The functions f and g are defined by the given equations, where $x \geq 0$:

$$f(x) = 33(0.4)^{x+3}$$

$$g(x) = 33(0.16)(0.4)^{x-2}$$

Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Answer to Question 6

Correct Answer: B

Question 7

Each of the following frequency tables represents a data set.
Which of these frequency tables represents the data set with the smallest standard deviation?

Value	Frequency (A)	Frequency (B)	Frequency (C)	Frequency (D)
5	0	11	15	18
6	10	17	15	14
7	55	19	15	11
8	10	17	15	14
9	0	11	15	18

- A. A
- B. B
- C. C
- D. D

Answer to Question 7

Correct Answer: A

Question 8

In $\triangle ABC$, $AB = 4$ and $BC = 5$.

If $\angle ABC < 90^\circ$, which of the following could be the area of $\triangle ABC$?

- A. 5
- B. 10
- C. 15
- D. 20

Answer to Question 8

Correct Answer: A

Question 9

The table summarizes the distribution of heights, in inches (“in”), for 300 participants in a study.

If a participant that is at most 71 inches tall is chosen at random, what is the probability that the participant was selected from Group 2?

	48–59 in.	60–71 in.	72–83 in.	Total
Group 1	20	60	20	100
Group 2	10	75	15	100
Group 3	30	45	25	100
Total	60	180	60	300

- A. $\frac{1}{4}$
- B. $\frac{17}{48}$
- C. $\frac{17}{60}$
- D. $\frac{17}{20}$

Answer to Question 9

Correct Answer: B

Question 10

The function f is defined as

$$f(x) = x^2 + ax + b$$

where a and b are positive integers.

If the minimum value of $f(x)$ is 7, which is a possible value of b ?

- A. 6
- B. 7
- C. 8
- D. 9

Answer to Question 10

Correct Answer: C

Question 11

The average annual energy cost for a certain home is \$4,334.

The homeowner plans to spend \$25,000 to install a geothermal heating system.

The homeowner estimates that the average annual energy cost will then be \$2,712.

Which of the following inequalities can be solved to find t , the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

A. $25,000 > (4,334 - 2,712)t$

B. $25,000 < (4,334 - 2,712)t$

C. $25,000 - 4,334 - 2,712t$

D. $25000 > \frac{4332}{2712}t$

Answer to Question 11

Correct Answer: B

Question 12

For the equation, b and c are constants.

For any real number t , the point

$$\left(\frac{-2t + 4}{3}, t \right)$$

lies on the graph of the equation

$$bx + 3y = c$$

What is the value of c ?

- A. -2
- B. 4
- C. 6
- D. 12

Answer to Question 12

Correct Answer: C

Question 13

A circle in the xy -plane has equation

$$(x - 2.5)^2 + (y - 3.5)^2 = 225$$

Line p is tangent to the circle at point $(-6.5, C)$, where C is a constant. Which of the following could be the slope of line p ?

- I. $-\frac{3}{4}$
 - II. $\frac{3}{4}$
 - III. $\frac{4}{3}$
- A. I only
- B. I and II only
- C. I and III only
- D. I, II, and III

Answer to Question 13

Correct Answer: B

Question 14

The function f is defined by

$$f(x) = a^{x-b}$$

where a and b are constants.

In the xy -plane, the graph of $y = f(x)$ passes through the points $(c, 7)$ and $(2c, 117)$, where c is a constant.

What is the value of b ?

- A. 4
- B. 17
- C. 110
- D. 124

Answer to Question 14

Correct Answer: A

Question 15

The given equation, where t is a positive constant, defines a circle in the xy -plane.

The radius of this circle is $\sqrt{41}$.

What is the value of t ?

$$2x^2 + 2y^2 + tx + ty - 78 = 0$$

- A. 1
- B. 4
- C. 7
- D. 8

Answer to Question 15

Correct Answer: B